Ming-Zhe Chong

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Lanzhou, China | Peking University, Beijing, China

Education Background

- 2021.9 - now **Peking University** (**PKU,** School of Electronics, **Supervisor: Prof. Ming-Yao Xia**)
  Major: Electromagnetic Field and Microwave Technology **PhD Candidate**
- 2025.5 - 2025.8 **City University of Hong Kong** (**CityUHK,** State Key Laboratory of Terahertz and Millimeter Waves)
  Research Assistant
- 2017.9 - 2021.6 **Huazhong University of Science and Technology** (**HUST,** School of Optical and Electronic Information)
  Major: Optoelectronic Information Science and Engineering **Bachelor of Engineering**
- 2019.9 - 2020.1 **Zhejiang University** (**ZJU,** School of Optical Science and Engineering) **Exchange Student**

Research Interests and Publications

Research Interests: **Terahertz (THz), Metasurfaces, EM wave manipulation**

1. Diffractive neural networks: Based on the diffraction propagation features between multiple layers of metasurfaces, diffractive neural network models are trained to achieve functions such as **non-reciprocal imaging**;

2. Terahertz communication: We design terahertz OAM mode-division-multiplexing wireless links, including mode conversion and filtering functions, and verify the large capacity characteristics of the terahertz channels;

3. Terahertz beam manipulation: The metasurfaces are designed and fabricated, enabling the realization of various terahertz structured beams such as focusing arrays, OAM combs, and surface Bessel beams.

Publications: I have **7** SCI papers accepted or published as the (co-)first author; **1** paper published as the first author in an EI conference.

SCI papers:

Ming-Zhe Chong, et al. "Orbital angular momentum processing via diffractive networks for terahertz mode-division-multiplexed wireless links." ***Light: Science & Applications***, **2026**, accepted. (**IF=23.4**)

Ming-Zhe Chong, et al. "Janus meta-imager: asymmetric image transmission and transformation enabled by diffractive neural networks." ***PhotoniX***, **2025**, 6(1): 60. (**IF=19.1**)

Ming-Zhe Chong, et al. "Reconfigurable terahertz beam splitters enabled by inverse-designed meta-devices." ***Photonics Research***, **2025**, 13(8): 2328-2338. (**IF=7.2**)

Ming-Zhe Chong, et al. "Generation of polarization-multiplexed terahertz orbital angular momentum combs via all-silicon metasurfaces." ***Light: Advanced Manufacturing***, **2024**, 5(3): 400-409. (**IF=10.6**)

Ming-Zhe Chong, et al. "Spin-decoupled excitation and wavefront shaping of structured surface waves via on-chip terahertz metasurfaces." ***Nanoscale***, **2023**, 15(9): 4515-4522. (**IF=5.1**)

Ming-Zhe Chong, et al. "Nonlinear modulation of terahertz waves based on a MAPbI3/Gold/Si hybrid plasmon-induced transparency (PIT) metasurface." ***Optical Materials***, **2022**, 129: 112554. (**IF=4.2**)

Awards and Honours

- 2025 PKU **BYD Scholarship**
- 2025 PKU **Award for Scientific Research**
- 2024 PKU **Merit Student**
- 2024 PKU **"Challenge Cup"**:
  Special Prize of the Youth Science Award
- 2021 HUST **Outstanding Graduate**
- 2018 **National Mathematics Competition**
  for College Students: **First Prize**

Other

- **Skills:** 1. Proficient in **MATLAB** and **Python** programming;
  2. Familiar with graphic design software,
  such as **Adobe Illustrator**;
  3. Proficient in electromagnetic simulation software,
  such as **CST**;
  4. Familiar with **terahertz experiment**,
  be able to build and operate THz testing systems.
- **English:** **IELTS** (Academic) **7.0**
- **Hobbies:** Mountaineering, hiking, badminton, etc.